2016 ICM Problem F
Modeling Refugee Immigration Policies

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Abstract

The European migrant crisis is an ongoing event involving the mass departure of citizens from Muslim-majority countries, primarily in Western Asia and Africa, towards Europe. The majority of the refugees fleeing towards European Union countries are doing so to avoid war in their native countries and persecution. According to the United Nations High Commissioner for Refugees, 57% of refugees are adult males, 17% are adult females, and 26% are children (India, 2015). Unfortunately, the poverty and inherent panic of the refugees has forced them to undergo poor travelling conditions. Many travelling by land must walk along dangerous highways, and those travelling by sea often use unsafe dinghies and rafts. This has resulted in the death of more than 5 000 refugees in 2016, as well as the collapse of five boats carrying over 2 000 refugees in April 2015, resulting in 1 200 deaths (Quinn, 2016).

Background

As hundreds of thousands refugees arriving europe everyday, their first entry points are not limited to Greece, Italy and Hungary, but as well as Romania, such Eastern part of europe. However, the massive number of refugees through Greece, Italy and Hungary had drawn an attention of other countries in Europe. At 2016, most members of European Union are agreed to take responsibilities to relocate those refugees. Although "huge numbers have been applying for asylum, the number of people being given asylum is far lower" (Migrant crisis, 2016). Others who do not get a approval, still stay illegal in one of those safe country, we defined as heaven. They may tend to move around to find a country that they may live for a while. However, "most died on the crossing from north Africa to Italy, and more than 800 died in the Aegean crossing

from Turkey to Greece" (BBC). The most reason causing their death are sickness and lack of foods. Some routes from Mediterrane are considered extremely dangerous due to the uncertainty of the ocean since they have to travel by boat, especially from Libya to Italy or Greece since the only way to get out is crossing the Ocean, which seems like that they do not have any other choices. In the case of a sharp increase in the number of refugees, the EU's response become weakness and fragmentation among member states, lack of solution of the refugee problem, and finally led to a crisis situation.

The Refugee Crisis

According to The 1951 Refugee Convention, refugees are people escape from their country because of the fear of being persecuted in their country (Sengupta, 2015). Asylum-seekers are people not claiming themselves as refugees but asking for international protections (UNHCR, 2017), it is hard to distinguish these 2 types of people as refugees can claim for asylum and asylum-seekers can declare as refugees if they can (BBC, 2016). Thus, this project assumes asylum-seekers as refugees for simplicity.

The goal of this study is to help develop a better understanding of the factors involved with facilitating the movement of refugees from their countries of origin into safehaven countries, and to provide potential solutions to the myriad of problems that have arisen. Specifically, to create an optimal network flow to maximize the number of refugees in designated "safe" countries, or to minimize the movement of refugees across Europe with the limitations each country has. There are six specific tasks which need to be accomplished:

1. Determine all factors that could influence the movement of refugees
2. Create an optimal network flow of refugee movement
3. Discuss any factors included in the model which are subject to change and what effect they might have
4. Write a report of the model and propose policies to support it

5. Discuss any external factors which might have an effect on the model
6. Discuss the effectiveness of the model on a much larger scale

Metrics of the Refugee Crisis

To choose a relevant objective function, we need to account for the accuracy of the solutions compared to the reality. If we maximize the number of refugees getting into safehaven countries, there is a really wide gap between safe and dangerous, which is very hard to achieve a meaningful result. Whereas minimizing the movement of refugees through europe is much easier to formulate and the solutions are able to reflect the reality. So the minimum flow problem suits the problem's solvability more.

There exist numerous factors which in reality affect the movement of refugees: refugee age, health, economic wealth, level of education, and languages spoken all influence the probability of a refugee being accepted into a given country. However, due to time constraints, budget constraints, and difficulty of estimating all of these factors, they have been excluded from the model; all refugees are viewed equally. Factors related to the safehaven countries such as transportation routes and methods, and the respective country's capacity have been factored into the model by way of assigning a relative cost to each travel route. Also, a quota has been assigned to each country, but quota for a country is quite subjective, different stakeholders may set different amount of quota for their own countries. In order to get an appropriate quota for each country, it is better to use some unbiased parameters such as the wealth of the country and its population density.

The definitions of parameters we are going to use :

Source Countries in the set S:

- Turkey
- Morocco(Western Mediterranean)
- Libya(Central Mediterranean)
- Serbia (West Balkans)
- Ukraine (Eastern Borders)

These are the main source countries, they give out refugees by assuming that in a constant rate k_s, so all these refugees are distributed to the following receiving countries in the set A :

- Spain
- Germany
- France
- Sweden
- UK
- Czech Republic
- Austria
- Hungary
- Poland
- Switzerland
- Belgium
- Italy
- Ireland
- Romania
- Bulgaria
- Greece
- Etc… smaller countries are not included due to the size limitation of the programming software we are using.

Quota Q_i

Every receiving country has its quota determined by using the country's population, the GDP of the country, the total GDP of European Union countries, and the total number of refugees that entered Europe in 2016. So the country with higher GDP has higher quota.

Objective function

The objective function is going to count the total cost all moving refugees have paid for the routes over a particular month. So the model is trying to minimize the objective function, that is to eliminate unnecessary routes as well as decide choices of routes.

Routes $e_{i,j}$

The route variable is independent, it contains a value, that is the number of refugees passing that route from country i to country j. So back to the objective function, this variable means the number of refugees paying $C_{i,j}$ in that month. The source countries have special properties on the routes, where the second and third constraints are showing. The second constraint implies that the total number of refugees getting out of a source country s has to be the constant k_s, while returning to country s is not allowed, so the third constraint has all such routes set to zero.

The constant rate k_s is actually computed according to the number of refugees taking that starting route in 2015, and divided by 12 months (Frontex, 2017). Here we are assuming no exogenous events affecting the flow.

Cost $C_{i,j}$

The cost for each route depends on its distance, available transportations, and the border controls. In fact, this cost is paid by the refugees, so it has to be correlated to the workload of the refugees. The value of the cost from country i to country j is the travelling distance of the shortest route measured from Google Map (Google Map). Most of the time, refugees were walking to their destination, especially when they are crossing the border illegally, unless they take the sea route (Warren, 2015). Moreover, for instance a refugee takes train from Germany to France, is much quicker than on foot, but he or she has paid for the ticket, which is usually equivalent to the traveling distance, the price of the ticket should also be taken into the objective function for minimization. That is why we use distance as the cost instead of

time with the consideration of which transportations are taken. Since our model sets the time interval as monthly, refugees probably have enough time to move from a country to another, even to the furthest one, (Google Map) so we don't need to restrict the long route by time.

Appendix B1 and B2 show how the routes are like, a route can only be a choice if it cross exactly one border, that is saying the countries are adjacent, others are assigned with a sufficiently large cost so that no refugees will take them.

<u>The population of refugees</u> X_i^n

X_i^n is a dependent variable, it depends on the flow of refugees across country i and the previous number of refugees in that country by the conservation of network flow, that is the first constraint of the problem. Together with the quota given to each country, the fourth constraint restricts this variable exceeding the quota.

<u>Solving the problem by using AMPL</u>

The problem is written as codes in AMPL language, and solved as an integer programming problem by CPLEX solver. See Appendix A1,2,3. The results are in Appendix B3, C1,...,12. B3 displays the value of the objective function on the first row, next are the number of refugees X_i^n for each country across each month and lastly the percentage of refugees comparing to each country's quota. The result shows the optimal value grows gradually by each month, which is reasonable because refugees are first come first serve.

Flow of Refugees

The dynamics of refugee crisis is actually quite obvious and understandable, the late arriving refugees must take a higher cost route, that is to go through more countries. The directed adjacency matrices in Appendix C1 to C12 show the flow of refugees numerically, but that is still very hard to observe what has actually happened. Appendix D1 to D12 are maps

illustrating the flow according to the matrices together with the percentage of refugees from the given quota in each month. This clearly shows that refugees are filling up the closest countries first, until they are full, then move on to the next closest ones. Although the greedy algorithm or the minimum spanning tree algorithm are kind of doing the same approach, unfortunately there are multiple sources which may interfere each other or form cycles, furthermore these algorithms may not work efficiently with our source countries' constraints and the quota constraints.

Luckily, it is not hard to see that some routes are always there after some time, i.e. some countries adjacent to the sources are also functioning like a source country, this implies that we can reduce number of variables and convert the problem into a smaller one by replacing the source country.

However, if we look back to the numbers on the flow, there is a huge flow of refugees through exactly some countries, for instance Hungary, because of the geographic of Europe. This may bring along some overcrowd problems, such as public hygiene, virus outbreak, and shelters' condition.

Dynamics of the Crisis

The EU (European Union) Common European Asylum System (CEAS) allows EU members "[to] choose to return asylum seekers to their country of first entry to process their asylum claim, so long as that country has an effective asylum system." As a result, this system works unfairly for countries that are located in the south side of Europe, such as Greece and Italy, where the most refugees arrive. The most important thing to be considered is health issue. As WHO suggest, given a health check for every refugees upon their arrival, then divide them into different groups. Also, consider human rights to be equally distributed among all refugees, regardless of their age, sex, and other factors. Since the relocation scheme has been approved in Europe Union, each member has agreed to help settle incoming refugees based on their

quotas. “UNHCR and NGOs recognize the importance of strengthening partnerships to identify refugees in need of solutions, including resettlement, and they acknowledge the need for balanced protection delivery among regions and within countries, including urban, camp and other operational contexts. “(NGO Toolkit for Practical Cooperation on Resettlement). In 2015, “Some European states, led by Germany, recognized that their strategy of seeking to block refugees moving across borders was unrealistic and harmful.” (The EU Has a Solidarity Crisis. Not a Refugee Crisis, 2017). However, it’s hard for refugees to choose other destinations, such as Canada, China, etc. For example, to come to China need a visa and take airplane , however apply refugee visa outside a country is very difficult. In this situation, the potential refugees need UNHCR's recommendation with few quotas. Without valid visa, refugees could not get on a airplane to security regulation. In the end, Europe is the most realistic choice for most refugees.

Policy to Support Refugee Model

It is clear to see that Greece and Italy are the first entry of most refugees are chosen, the”hotspots” in those country are overcrowded and it brings up the new issue that how to make a sustainable and affordable asylum system for EU. First approach is to set more refugee camp in the Middle East and Africa, where refugee could make applications overseas. This way avoid the dangerous experience by refugees who traveled across the Mediterranean in small boats. Also by increase cooperation with source countries , including economic aid and health assistant in Turkey, Lebanon and Jordan and coordination with Saudi Arabia, Iran and other countries on regional political and security affairs. Next option is to help the refugees to settle or transit to neighbour country by enhance common border line at the same time. Also looked the system in Canada, Canadian refugee system has two main parts to help refugee application from both outside and inside of Canada. “Resettled refugees get initial assistance from either the federal government, the Province of Quebec, or private sponsors (organizations or groups of people in Canada).” (The refugee system in Canada, 2016). Canada has a history working with

private sponsorship program to help refugees from Syrian by "providing the newcomers with basic material needs like food, clothing, and housing, as well as help adjusting to Canadian society" (The Unique Programme that brings refugees to Canada, 2016). Also, the local food bank helps to interview those refugees to determine the specific needs for them. On September 20th, 2016, Canadian government announces a new program that implement with worldwide private sponsorship along with UN Refugee Agency and Open Society Foundation to create new program that aim to supporting the refugees crisis in EU. On the other hand, the growing security problems associated with refugees has effect on the internal security environment in Europe and the political environment for refugee problems. EU has proposed the establishment of a standing border force and a coast guard to replace the existing functions of the EU Border Agency, in order to strengthen the common border control of the EU countries.

Exogenous Events

There are many changes within year, for stance in 2014 Bulgaria had set up fences restricting refugees from Turkey to get in, this led to the majority of the refugees from Turkey taking the sea route through Aegean Sea to Greece, some refugees even broke the fences. (Al Mukhtar, 2015) Border control policy certainly increase the cost of that route, it slows down the flow and refugees have to take a more difficult route instead.

After the terrorist attacks in Paris and in Berlin, not just the European, all the countries around the world are concerning about the refugee crisis.

Scalability

Unfortunately, the model does not hold up when the problem is scaled up by a factor of ten. This is to be expected though, as realistically the European Union wouldn't hold up well faced with the task of taking in more than eighteen million refugees in a year. Currently, the

majority of EU countries have more refugees than their quotas allow for; the only solution to this issue would be to have every country in the EU (and ideally countries out of the EU) take in more refugees than they currently are, but this would only increase the strain on local economies and aiding organizations to keep the refugees safe. An increase in the number of refugees seeking asylum would slow down the whole process and inevitably lead to an increase in mortalities. Not only would transportation be slower, but all bureaucratic processes (especially applications for asylum) would slow to a crawl with such an influx, considering the application rate for asylum has reached as high as 60 000 per month from Syrian refugees (UNHCR Syria Regional Refugee Response, 2016).

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Appendix

A1.

simple.dat | *refugee.mod | refugee.run

```
data;

set S := Libya  Morocco Serbia  Turkey  Ukraine;
set A := Austria    Belgium Bulgaria    Czech   France  Germany Greece  Hungary Italy   Netherla

param entering := Libya 12828   Morocco 597 Serbia  63742   Turkey  73782   Ukraine 1034;

param X_prev := Austria 0       Belgium 0       Bulgaria 0
Czech   0       France  0       Germany 0       Greece  0
Hungary 0       Italy   0       Netherlands 0   Poland  0
Romania 0       Spain   0       Sweden  0       Switzerland 0   UK  0;

param quota :=
Austria 39194
Belgium 47394
Bulgaria    5158
Czech   34777
France  268085
Germany 383650
Greece  20990
Hungary 13394
Italy   218779
Netherlands 86512
Poland  107944
Romania 44024
Spain   168311
Sweden  49468
Switzerland 58013
UK  278140;

param cost: Austria Belgium Bulgaria    Czech   France  Germany Greece  Hungary Italy   Netherla
Austria 999999  999999  999999  368 999999  688 999999  470 936 999999  999999  999999  999999
Belgium 999999  999999  999999  999999  660 540 999999  999999  999999  230 999999  999999  9999
Bulgaria    999999  999999  999999  999999  999999  999999  791 999999  999999  999999  999999
Czech   368 999999  999999  999999  999999  507 999999  999999  999999  999999  453 999999  9999
France  999999  660 999999  999999  999999  999999  999999  999999  999999  999999  999999  9999
```

A2.

simple.dat | *refugee.mod | refugee.run

```
set S;      # sources
set A;     # countries receiving refugees

param X_prev {A} >= 0;
param entering {S} >= 0;
param cost {A union S, A union S} >= 0;
param quota {A} >= 0;

var X {i in A} integer >= 0;
var E {i in A union S, j in A union S} integer >= 0;

minimize Total_Cost:
   sum {i in A union S, j in A union S} cost[i,j] * E[i,j];

subject to Conservation {i in A}:
   X[i] - sum {j in A union S} E[j,i] + sum {j in A union S} E[i,j] = X_prev[i];

subject to Sources {s in S}:
    sum {j in A} (E[s,j]) = entering[s];

subject to Quotaing {i in A} :
    X[i] <= quota[i];
```

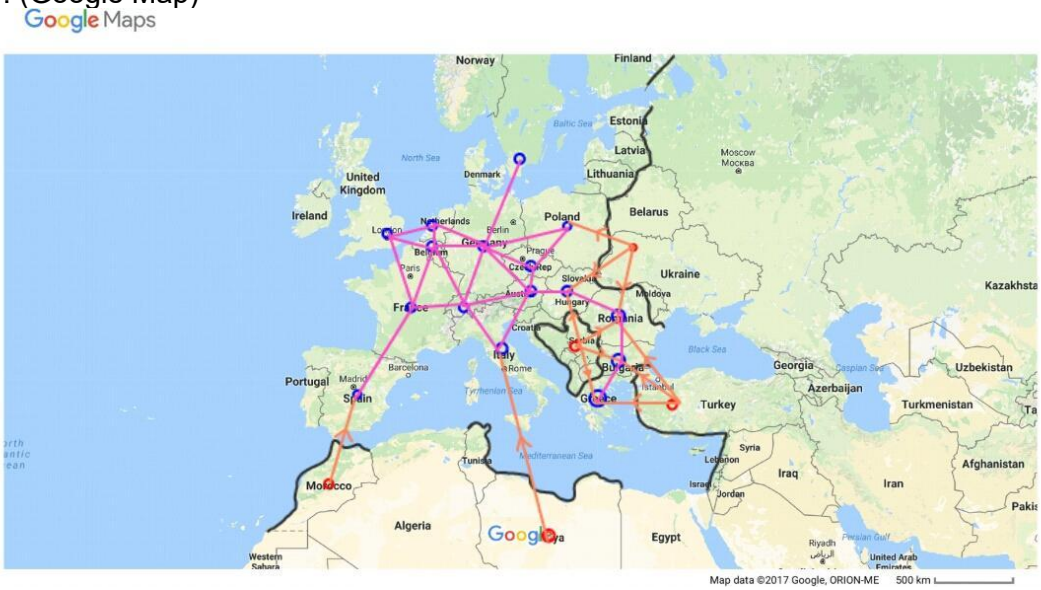
A3.

simple.dat | *refugee.mod | refugee.run

```
model refugee.mod;
data simple.dat;
option solver cplex;

for {n in 1..12} {
       display n;
       solve;
       display X;
       display E;
        for {i in A}{
            let X_prev[i] := X[i];
            }
    }
```

B1. (Google Map)



B2. (Google Map)

Distance in KM	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlands	Poland	Romania	Spain	Sweden	Switzland	UK	Libya	Morocco	Serbia	Turkey	Ukraine
Austria	999999	999999	999999	368	999999	688	999999	470	936	999999	999999	999999	999999	999999	694	999999	999999	999999	999999	999999	999999
Belgium	999999	999999	999999	999999	660	540	999999	999999	999999	230	999999	999999	999999	999999	999999	402	999999	999999	999999	999999	999999
Bulgaria	999999	999999	999999	999999	999999	999999	791	999999	999999	999999	999999	515	999999	999999	999999	999999	999999	999999	999999	999999	999999
Czech	368	999999	999999	999999	999999	507	999999	999999	999999	999999	453	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
France	999999	660	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	943	999999	553	827	999999	999999	999999	999999	999999
Germany	688	540	999999	507	999999	999999	999999	999999	999999	467	760	999999	999999	1316	662	999999	999999	999999	999999	999999	999999
Greece	999999	999999	791	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Hungary	470	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	600	999999	999999	999999	999999	999999	999999	999999	999999	999999
Italy	936	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	837	999999	999999	999999	999999	999999	999999
Netherlands	999999	230	999999	999999	999999	467	999999	999999	999999	999999	999999	999999	999999	999999	999999	519	999999	999999	999999	999999	999999
Poland	999999	999999	999999	453	999999	760	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Romania	999999	999999	515	999999	999999	999999	999999	600	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Spain	999999	999999	999999	999999	943	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Sweden	999999	999999	999999	999999	999999	1316	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Switzland	694	999999	999999	999999	553	662	999999	999999	837	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
UK	999999	402	999999	999999	827	999999	999999	999999	999999	519	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Libya	999999	999999	999999	999999	999999	999999	999999	999999	1776	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Morocco	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	1010	999999	999999	999999	999999	999999	999999	999999	999999
Serbia	999999	999999	512	999999	999999	999999	756	433	999999	999999	999999	567	999999	999999	999999	999999	999999	999999	999999	999999	999999
Turkey	999999	999999	921	999999	999999	999999	1159	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999	999999
Ukraine	999999	999999	999999	999999	999999	999999	999999	1057	999999	999999	1051	693	999999	999999	999999	999999	999999	999999	999999	999999	999999

B3. (StatisticsTimes, 2016)

	Total Costs	0	182512282	355141302	361545462	361598916	373599946	394119614	429441778	457961112	470505243	521292614	540120798	599795798
Quota	# of refugees	x0	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10	x11	x12
39194	Austria	0	39194	39194	39194	39194	39194	39194	39194	39194	39194	39194	39194	39194
47394	Belgium	0	0	0	0	0	0	0	47267	47394	47394	47394	47394	47394
5158	Bulgaria	0	5158	5158	5158	5158	5158	5158	5158	5158	5158	5158	5158	5158
34777	Czech	0	14764	34777	34777	34777	34777	34777	34777	34777	34777	34777	34777	34777
268085	France	0	0	0	0	0	0	0	0	150352	268085	268085	268085	268085
383650	Germany	0	0	117511	255035	383650	383650	383650	383650	383650	383650	383650	383650	383650
20990	Greece	0	20990	20990	20990	20990	20990	20990	20990	20990	20990	20990	20990	20990
13394	Hungary	0	13394	13394	13394	13394	13394	13394	13394	13394	13394	13394	13394	13394
218779	Italy	0	12828	25656	38484	51312	64140	201172	218779	218779	218779	218779	218779	218779
86512	Netherlands	0	0	0	0	0	0	0	86512	86512	86512	86512	86512	86512
107944	Poland	0	1034	2068	3102	4136	93590	107944	107944	107944	107944	107944	107944	107944
44024	Romania	0	44024	44024	44024	44024	44024	44024	44024	44024	44024	44024	44024	44024
168311	Spain	0	597	1194	1791	2388	2985	3582	4179	4776	5373	5970	19395	168274
49468	Sweden	0	0	0	0	0	0	0	0	0	0	0	46364	49468
58013	Switzland	0	0	0	0	8909	58013	58013	58013	58013	58013	58013	58013	58013
278140	UK	0	0	0	0	0	0	0	0	907	34560	185946	278140	278140
		Percentage												
	Austria	0	100	100	100	100	100	100	100	100	100	100	100	100
	Belgium	0	0	0	0	0	0	0	99.73	100	100	100	100	100
	Bulgaria	0	100	100	100	100	100	100	100	100	100	100	100	100
	Czech	0	42.45	100	100	100	100	100	100	100	100	100	100	100
	France	0	0	0	0	0	0	0	0	56.08	100	100	100	100
	Germany	0	0	30.63	66.48	100	100	100	100	100	100	100	100	100
	Greece	0	100	100	100	100	100	100	100	100	100	100	100	100
	Hungary	0	100	100	100	100	100	100	100	100	100	100	100	100
	Italy	0	5.86	11.73	17.59	23.45	29.32	91.95	100	100	100	100	100	100
	Netherlands	0	0	0	0	0	0	0	100	100	100	100	100	100
	Poland	0	0.96	1.92	2.87	3.83	86.7	100	100	100	100	100	100	100
	Romania	0	100	100	100	100	100	100	100	100	100	100	100	100
	Spain	0	0.35	0.71	1.06	1.42	1.77	2.13	2.48	2.84	3.19	3.55	11.52	100
	Sweden	0	0	0	0	0	0	0	0	0	0	0	93.73	100
	Switzland	0	0	0	0	15.36	100	100	100	100	100	100	100	100
	UK	0	0	0	0	0	0	0	0	0.33	12.43	66.85	100	100

C1.

1	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria				14764												
Belgium																
Bulgaria												47634				
Czech																
France																
Germany																
Greece																
Hungary	53958															
Italy																
Netherlands																
Poland																
Romania								3610								
Spain																
Sweden																
Switzland																
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			52792				20990									
Ukraine											1034					

C2.

2	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria				20013		117511										
Belgium																
Bulgaria												73782				
Czech																
France																
Germany																
Greece																
Hungary	137524															
Italy																
Netherlands																
Poland																
Romania								73782								
Spain																
Sweden																
Switzland																
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C3.

3	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria						137524										
Belgium																
Bulgaria												73782				
Czech																
France																
Germany																
Greece																
Hungary	137524															
Italy																
Netherlands																
Poland																
Romania								73782								
Spain																
Sweden																
Switzland																
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C4.

4	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria						128615									8909	
Belgium																
Bulgaria												73782				
Czech																
France																
Germany																
Greece																
Hungary	137524															
Italy																
Netherlands																
Poland																
Romania								73782								
Spain																
Sweden																
Switzland																
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C5.

5	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria				88420											49104	
Belgium																
Bulgaria												73782				
Czech											88420					
France																
Germany																
Greece																
Hungary	137524															
Italy																
Netherlands																
Poland																
Romania								73782								
Spain																
Sweden																
Switzland																
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C6.

6	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria				13320					124204							
Belgium																
Bulgaria												73782				
Czech											13320					
France																
Germany																
Greece																
Hungary	137524															
Italy																
Netherlands																
Poland																
Romania								73782								
Spain																
Sweden																
Switzland																
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C7.

7	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria						132745			4779							
Belgium																
Bulgaria												73782				
Czech																
France																
Germany		47267								86512						
Greece																
Hungary	137524															
Italy																
Netherlands																
Poland						1034										
Romania								73782								
Spain																
Sweden																
Switzland																
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C8.

8	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria															137524	
Belgium																907
Bulgaria												73782				
Czech																
France																
Germany		1034														
Greece																
Hungary	137524															
Italy															12828	
Netherlands																
Poland						1034										
Romania								73782								
Spain																
Sweden																
Switzland					150352											
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C9.

9	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria						32619									104905	
Belgium																33653
Bulgaria												73782				
Czech																
France																
Germany		33653														
Greece																
Hungary	137524															
Italy															12828	
Netherlands																
Poland						1034										
Romania								73782								
Spain																
Sweden																
Switzland					117733											
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

C10.

10	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria						137524										
Belgium																138558
Bulgaria												73782				
Czech																
France																12828
Germany		138558														
Greece																
Hungary	137524															
Italy															12828	
Netherlands																
Poland						1034										
Romania								73782								
Spain																
Sweden																
Switzland					12828											
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

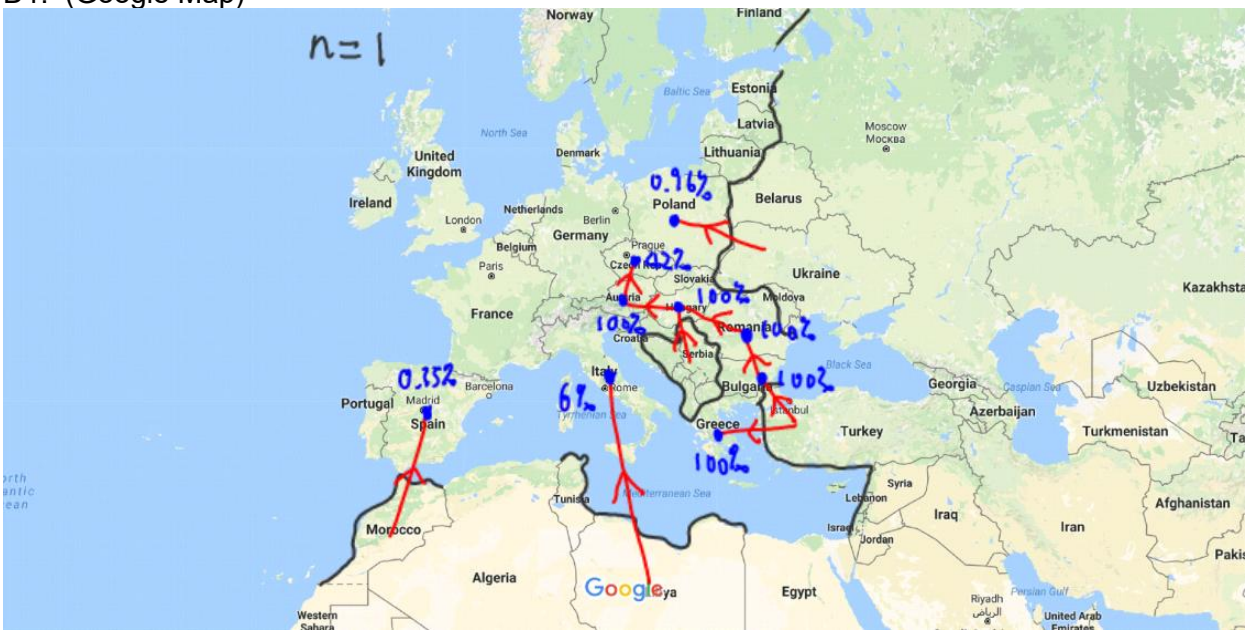
C11.

11	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria						137524										
Belgium																92194
Bulgaria												73782				
Czech																
France													12828			
Germany		92194												46364		
Greece																
Hungary	137524															
Italy															12828	
Netherlands																
Poland						1034										
Romania								73782								
Spain																
Sweden																
Switzland					12828											
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

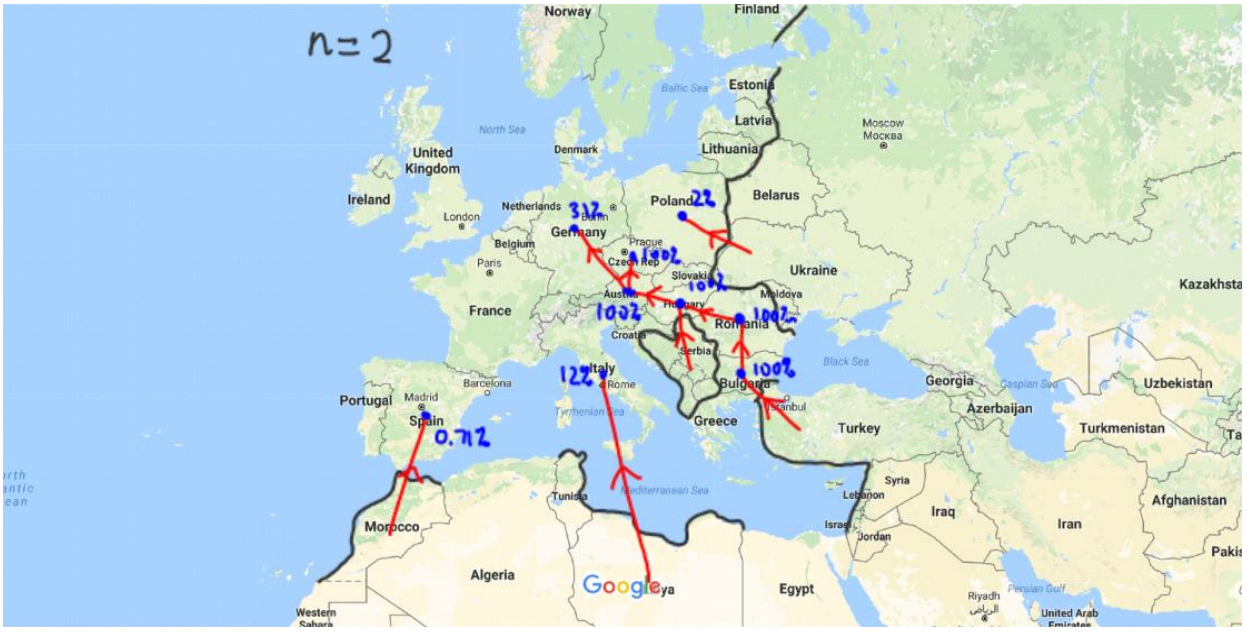
C12.

12	Austria	Belgium	Bulgaria	Czech	France	Germany	Greece	Hungary	Italy	Netherlanc	Poland	Romania	Spain	Sweden	Switzland	UK
Austria						2070									135454	
Belgium																
Bulgaria												73782				
Czech																
France													148282			
Germany														3104		
Greece																
Hungary	137524															
Italy															12828	
Netherlands																
Poland						1034										
Romania								73782								
Spain																
Sweden																
Switzland					148282											
UK																
Libya									12828							
Morocco													597			
Serbia								63742								
Turkey			73782													
Ukraine											1034					

D1. (Google Map)



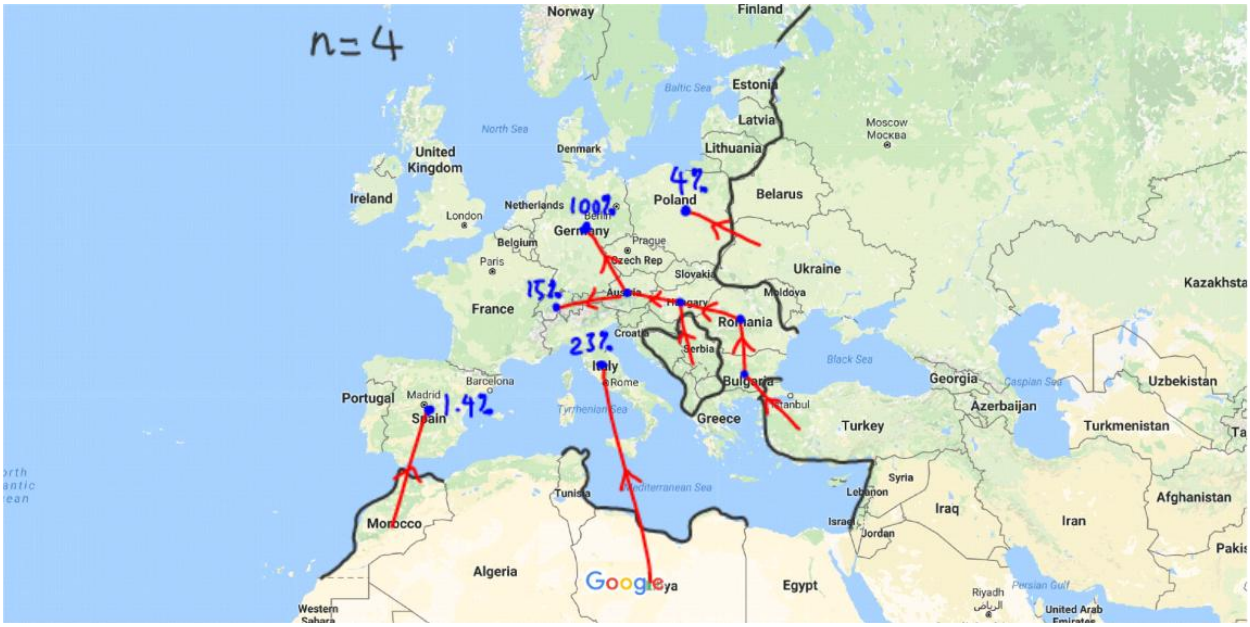
D2.



D3.

n=3
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
Berlin
66%
Germany
32
Poland
Belarus
Belgium
Paris
Prague
Czech Rep
Slovakia
Ukraine
Kazakhsta
Austria
Hungary
Moldova
France
Romania
Croatia
Serbia
Black Sea
18%
Italy
Rome
Bulgaria
Georgia
Caspian Sea
Uzbekistan
Portugal
Madrid
Barcelona
Spain
Tyrrhenian Sea
Greece
Istanbul
Azerbaijan
Turkey
Turkmenistan
Syria
Lebanon
Tunisia
Mediterranean Sea
Iraq
Afghanistan
Israel
Jordan
Iran
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab
Western

D4.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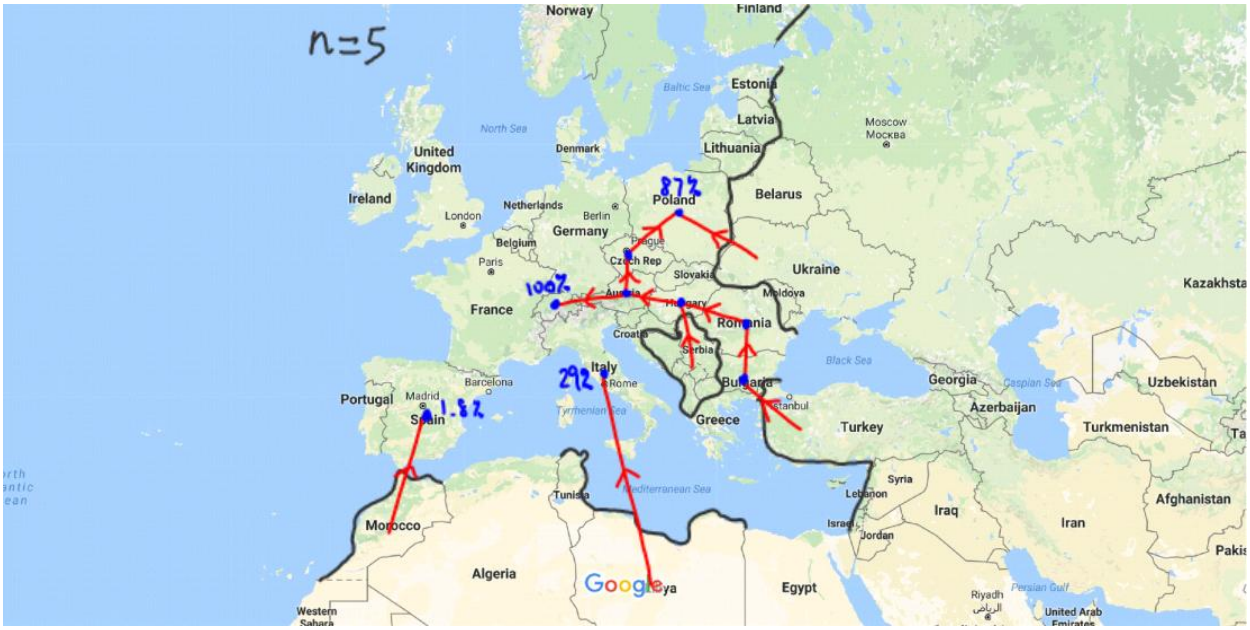
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
100%
Berlin
Germany
47
Poland
Belarus
Belgium
Paris
Prague
Czech Rep
Slovakia
Ukraine
Kazakhsta
15%
Austria
Hungary
Moldova
France
Romania
Croatia
23%
Serbia
Black Sea
Italy
Rome
Bulgaria
Georgia
Caspian Sea
Uzbekistan
Portugal
Madrid
Barcelona
Spain
1.4%
Tyrrhenian Sea
Greece
Istanbul
Azerbaijan
Turkey
Turkmenistan
Syria
Lebanon
Tunisia
Mediterranean Sea
Iraq
Afghanistan
Israel
Jordan
Iran
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab
Western

D5.


n=5
87%
100%
29%
1.8%
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
Berlin
Germany
Belgium
Paris
Poland
Belarus
Czech Rep
Slovakia
Ukraine
Austria
Hungary
Moldova
France
Romania
Croatia
Serbia
Italy
Rome
Barcelona
Portugal
Madrid
Spain
Bulgaria
Istanbul
Greece
Black Sea
Georgia
Caspian Sea
Azerbaijan
Uzbekistan
Turkmenistan
Turkey
Syria
Lebanon
Israel
Jordan
Iraq
Iran
Afghanistan
Tunisia
Mediterranean Sea
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab Emirates
Google

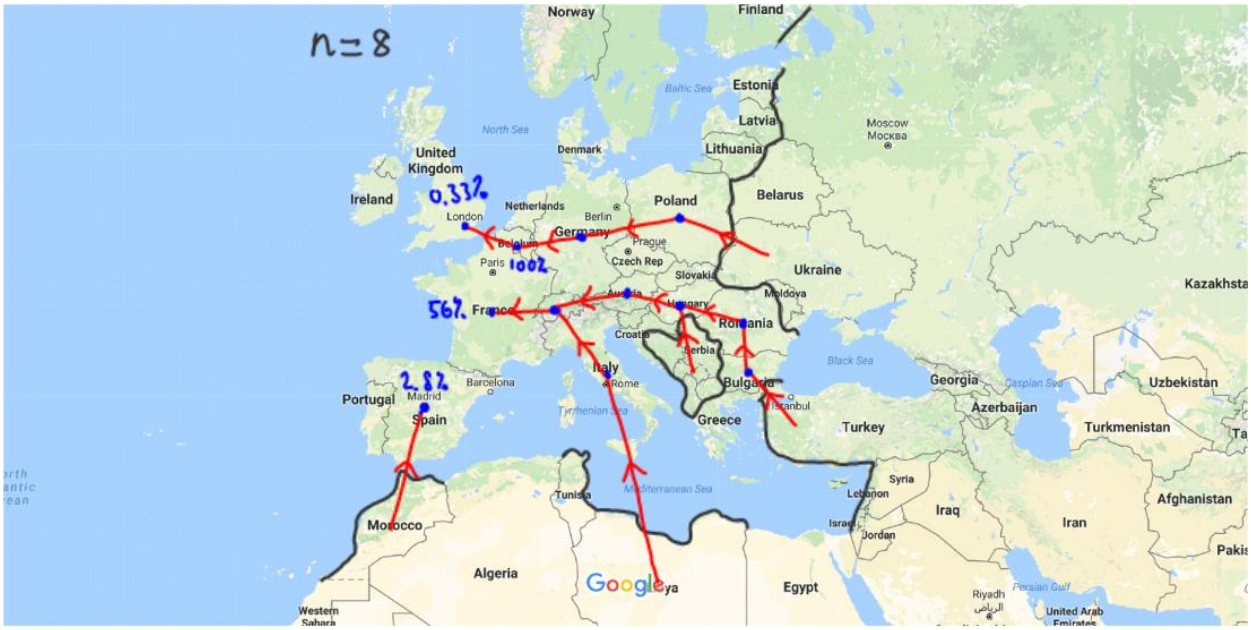
D6.

n=6
100%
92%
2.1%
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
Berlin
Germany
Belgium
Paris
Poland
Belarus
Czech Rep
Slovakia
Ukraine
Austria
Hungary
Moldova
France
Romania
Croatia
Serbia
Italy
Rome
Barcelona
Portugal
Madrid
Spain
Bulgaria
Istanbul
Greece
Black Sea
Georgia
Caspian Sea
Azerbaijan
Uzbekistan
Turkmenistan
Turkey
Syria
Lebanon
Israel
Jordan
Iraq
Iran
Afghanistan
Tunisia
Mediterranean Sea
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab Emirates
Google

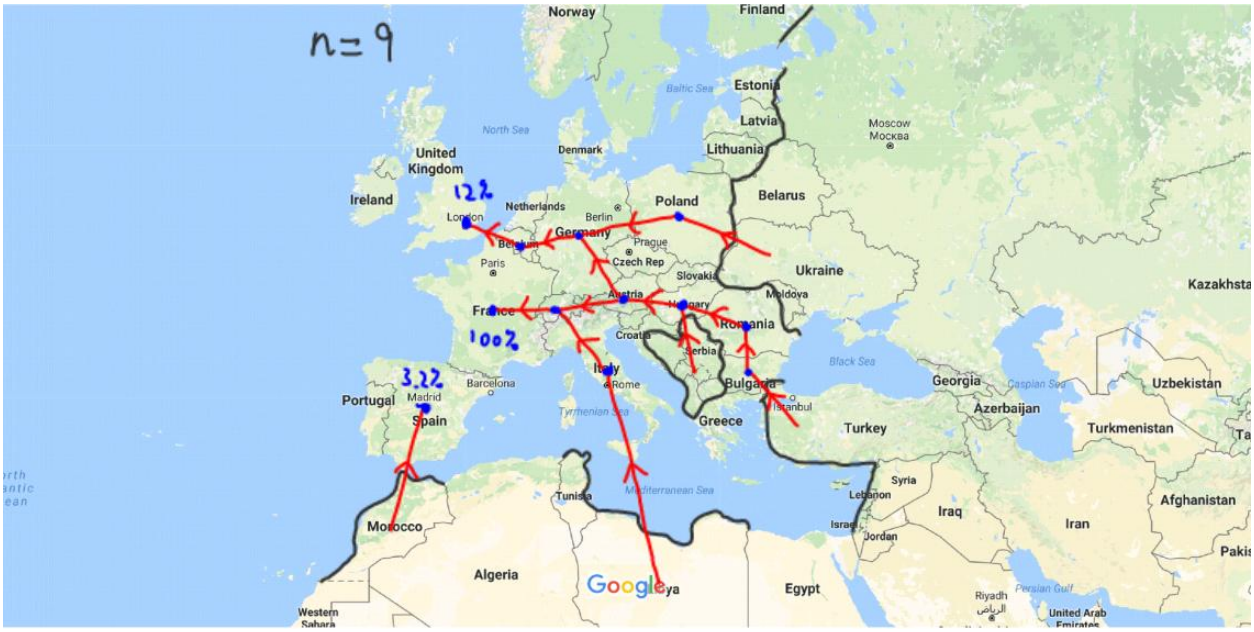
D7.

n=7
100%
99.7%
2.5%
100%
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
Berlin
Poland
Belarus
Germany
Prague
Czech Rep
Paris
Slovakia
Ukraine
Moldova
Austria
Hungary
France
Romania
Croatia
Serbia
Italy
Rome
Black Sea
Bulgaria
Portugal
Madrid
Barcelona
Spain
Istanbul
Greece
Georgia
Caspian Sea
Azerbaijan
Uzbekistan
Turkey
Turkmenistan
Syria
Lebanon
Tunisia
Mediterranean Sea
Iraq
Israel
Jordan
Afghanistan
Iran
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab
Western Sahara
Google

D8.


n=8
0.33%
100%
56%
2.8%
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
Berlin
Poland
Belarus
Germany
Prague
Czech Rep
Paris
Slovakia
Ukraine
Moldova
Austria
Hungary
France
Romania
Croatia
Serbia
Italy
Rome
Black Sea
Bulgaria
Portugal
Madrid
Barcelona
Spain
Istanbul
Greece
Georgia
Caspian Sea
Azerbaijan
Uzbekistan
Turkey
Turkmenistan
Syria
Lebanon
Tunisia
Mediterranean Sea
Iraq
Israel
Jordan
Afghanistan
Iran
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab
Western Sahara
Google

D9.


n=9
12%
100%
3.2%
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
Netherlands
Berlin
Poland
Belarus
London
Germany
Prague
Czech Rep
Paris
Slovakia
Ukraine
Moldova
Kazakhsta
Austria
Hungary
Romania
Croatia
Serbia
Italy
Rome
Black Sea
Bulgaria
Istanbul
Georgia
Caspian Sea
Uzbekistan
Portugal
Madrid
Barcelona
Spain
Tyrrhenian Sea
Greece
Azerbaijan
Turkey
Turkmenistan
Syria
Lebanon
Tunisia
Mediterranean Sea
Iraq
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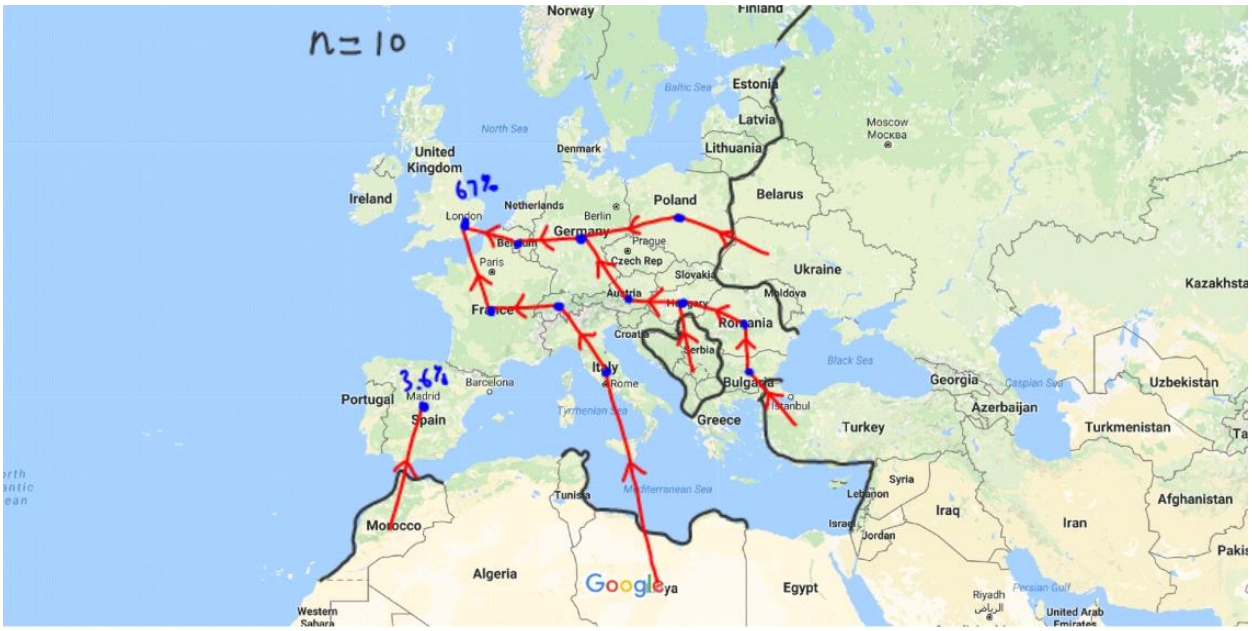
Jordan
Afghanistan
Iran
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab
Western Sahara

D10.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67%
3.6%
Norway
Finland
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Latvia
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Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
Netherlands
Berlin
Poland
Belarus
London
Germany
Prague
Czech Rep
Paris
Slovakia
Ukraine
Moldova
Kazakhsta
Austria
Hungary
Romania
Croatia
Serbia
Italy
Rome
Black Sea
Bulgaria
Istanbul
Georgia
Caspian Sea
Uzbekistan
Portugal
Madrid
Barcelona
Spain
Tyrrhenian Sea
Greece
Azerbaijan
Turkey
Turkmenistan
Syria
Lebanon
Tunisia
Mediterranean Sea
Iraq
Israel
Jordan
Afghanistan
Iran
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab
Western Sahara

D11.

n=11
94%
100%
13%
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
Berlin
Poland
Belarus
Germany
Prague
Czech Rep
Slovakia
Ukraine
Moldova
Austria
Hungary
Romania
France
Croatia
Serbia
Italy
Rome
Black Sea
Bulgaria
Istanbul
Georgia
Caspian Sea
Uzbekistan
Azerbaijan
Portugal
Madrid
Barcelona
Spain
Tyrrhenian Sea
Greece
Turkey
Turkmenistan
Syria
Lebanon
Tunisia
Mediterranean Sea
Iraq
Afghanistan
Israel
Jordan
Iran
Morocco
Algeria
Egypt
Riyadh
Persian Gulf
United Arab
Western Sahara
Google

D12.

n=12
100%
100%
Norway
Finland
Estonia
Latvia
Lithuania
Moscow
Москва
Baltic Sea
North Sea
Denmark
United Kingdom
Ireland
London
Netherlands
Berlin
Poland
Belarus
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Czech Rep
Slovakia
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Moldova
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France
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Morocco
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Egypt
Riyadh
Persian Gulf
United Arab
Western Sahara
Google